Sustainable Energy Technologies

Industries and industrial heat

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Study goals

Today, we want to understand

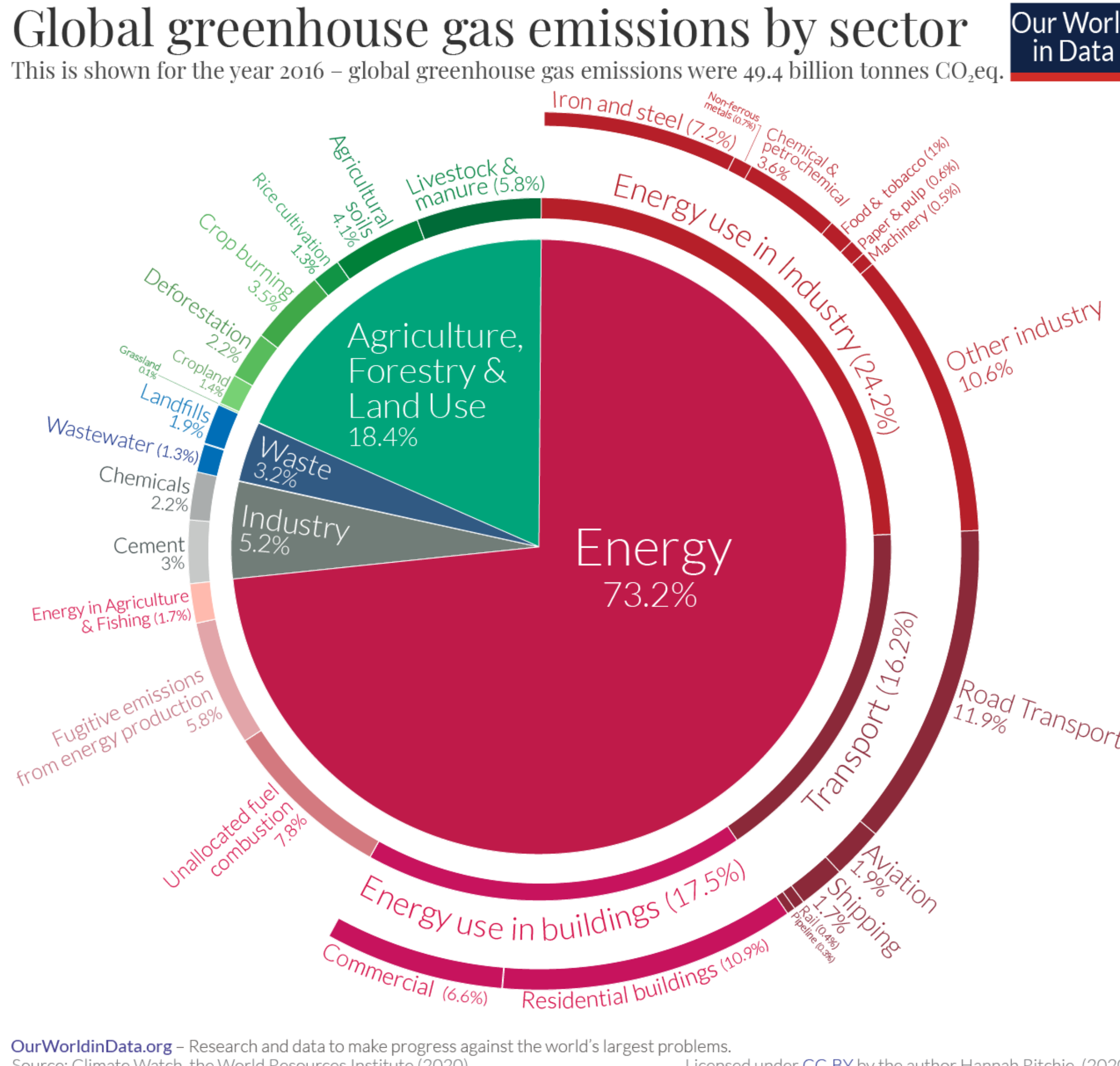
What are hard-to-abate sectors?

Understanding decarbonization strategies for steel

Understanding decarbonization strategies for cement

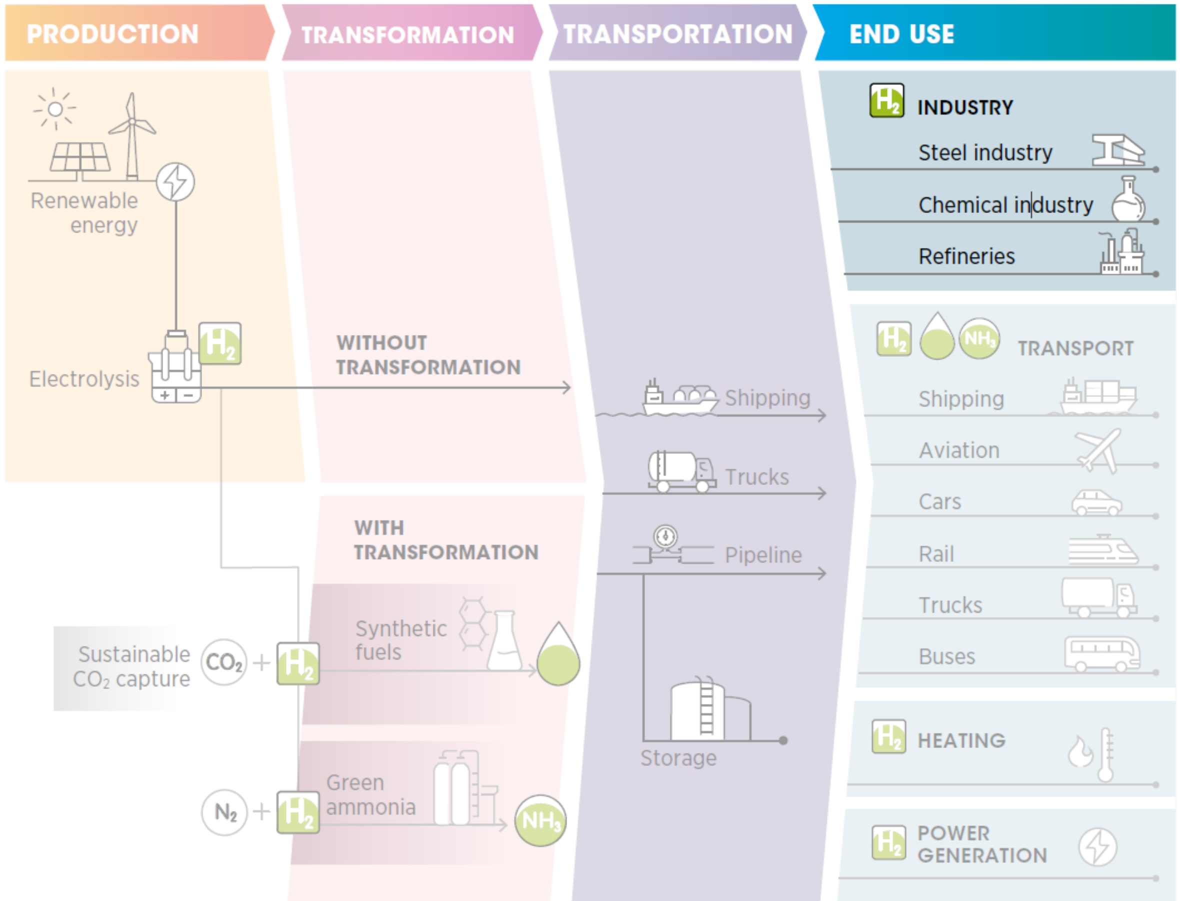
“Hard-to-abate” industries

Industry is a large emitter of GHG



Hard-to-abate industries

Industries (+ aviation/shipping covered earlier)



IRENA, 2022: Green hydrogen for industry - a guide to policy making

Steel making

Common misconceptions on green steel

(Understand why these statements are wrong, by the end of today)

1. "If we recycled steel, we would not need any new steel"
2. "Using electric arc cannot be done at the scale needed"
3. "Hydrogen for steel is unproven technology and might produce weak steel"
4. "Steel by definition has carbon; you cannot have green steel"

Extracting iron

Iron found in abundance on earth:

- Looks like red "Mars" dirt.
- But it comes bound to oxygen

Iron ore = Iron + oxygen in some shape:

- Magnetite Fe_3O_4
- Hematite Fe_2O_3

We want to extract the iron from that dirt

- By stripping out oxygen
- In chemistry, this is called to "reduce" a compound
- There are three (established) ways to make steel

Making steel

BF-BOF route

Coke = a form of coal

Coke reacts to **form** CO
CO **reduces** iron ore:

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Output: **Molten pig iron**
(~4–5% C <– too much)

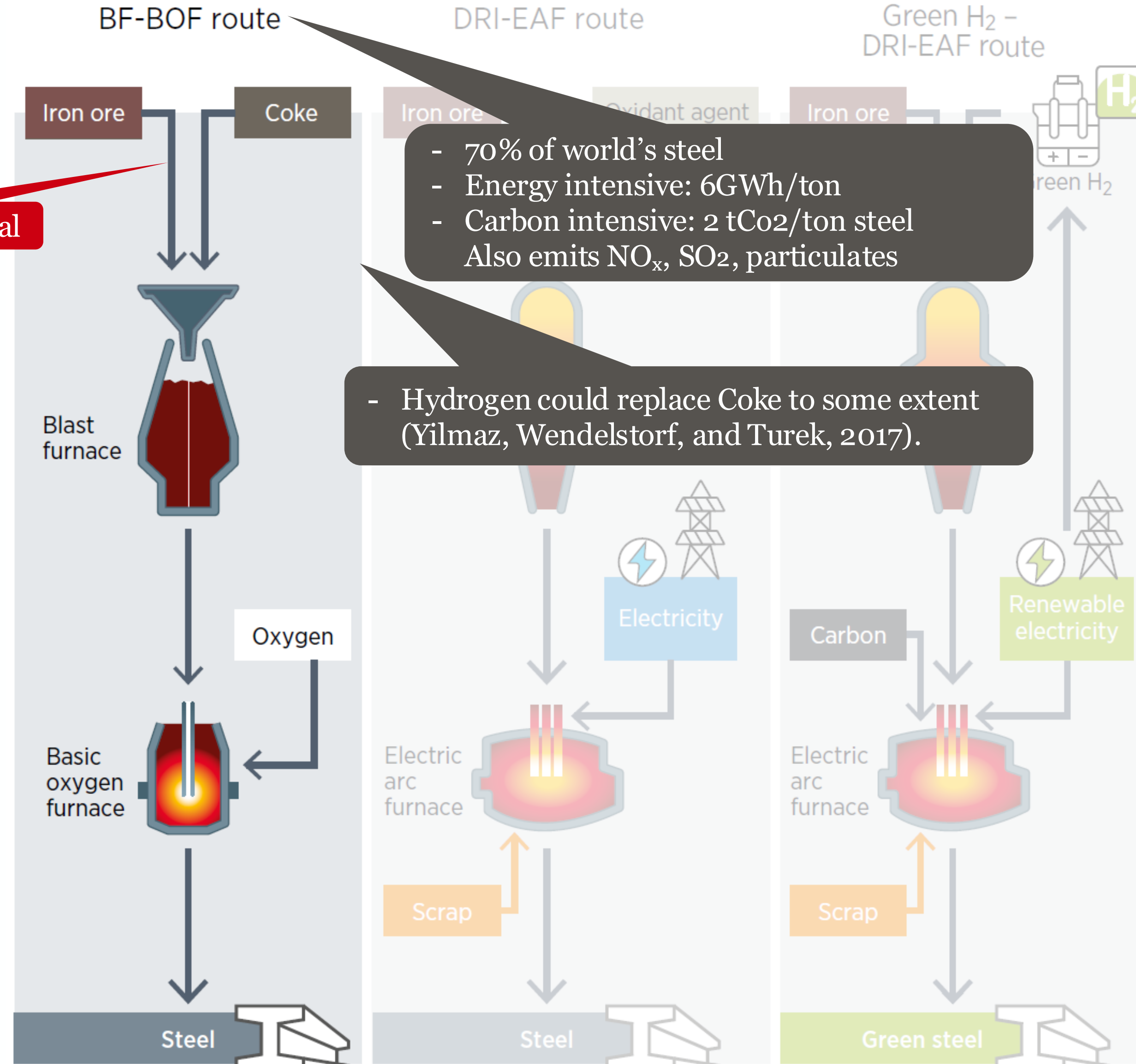
High-purity O_2 jet **oxidizes** impurities

Carbon **removed** as CO/CO_2:

$C + O_2 \rightarrow CO/CO_2$

Output: **Molten crude steel**

(~0.1–1.5% C)



- 70% of world's steel
- Energy intensive: 6GWh/ton
- Carbon intensive: 2 tCO_2/ton steel
  Also emits NO_x, SO_2, particulates

- Hydrogen could replace Coke to some extent (Yilmaz, Wendelstorf, and Turek, 2017).

Making steel

DRI-EAF

Iron ore **reduced to <u>solid</u>** iron (sponge iron/DRI)

Reductant: Syngas ($CO + H_2$) from natural gas

Reaction: $Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$

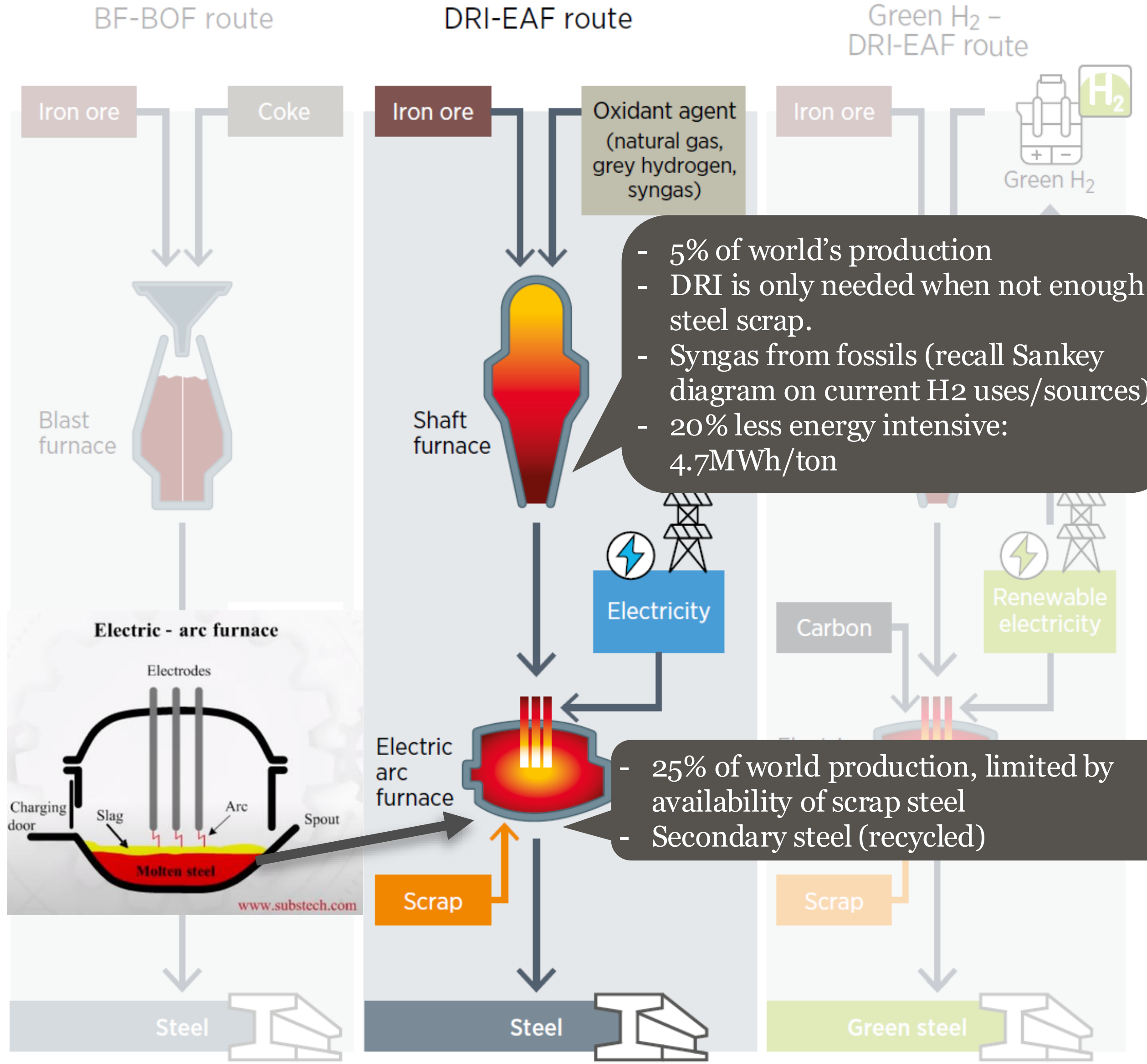
Output: **Direct Reduced Iron**

DRI and/or scrap **melted in Electric Arc** Furnace

High-voltage arcs generate intense heat

Alloying and fluxes added

Output: **Liquid steel**



- 5% of world's production
- DRI is only needed when not enough steel scrap.
- Syngas from fossils (recall Sankey diagram on current H2 uses/sources)
- 20% less energy intensive: 4.7MWh/ton

- 25% of world production, limited by availability of scrap steel
- Secondary steel (recycled)

Making steel

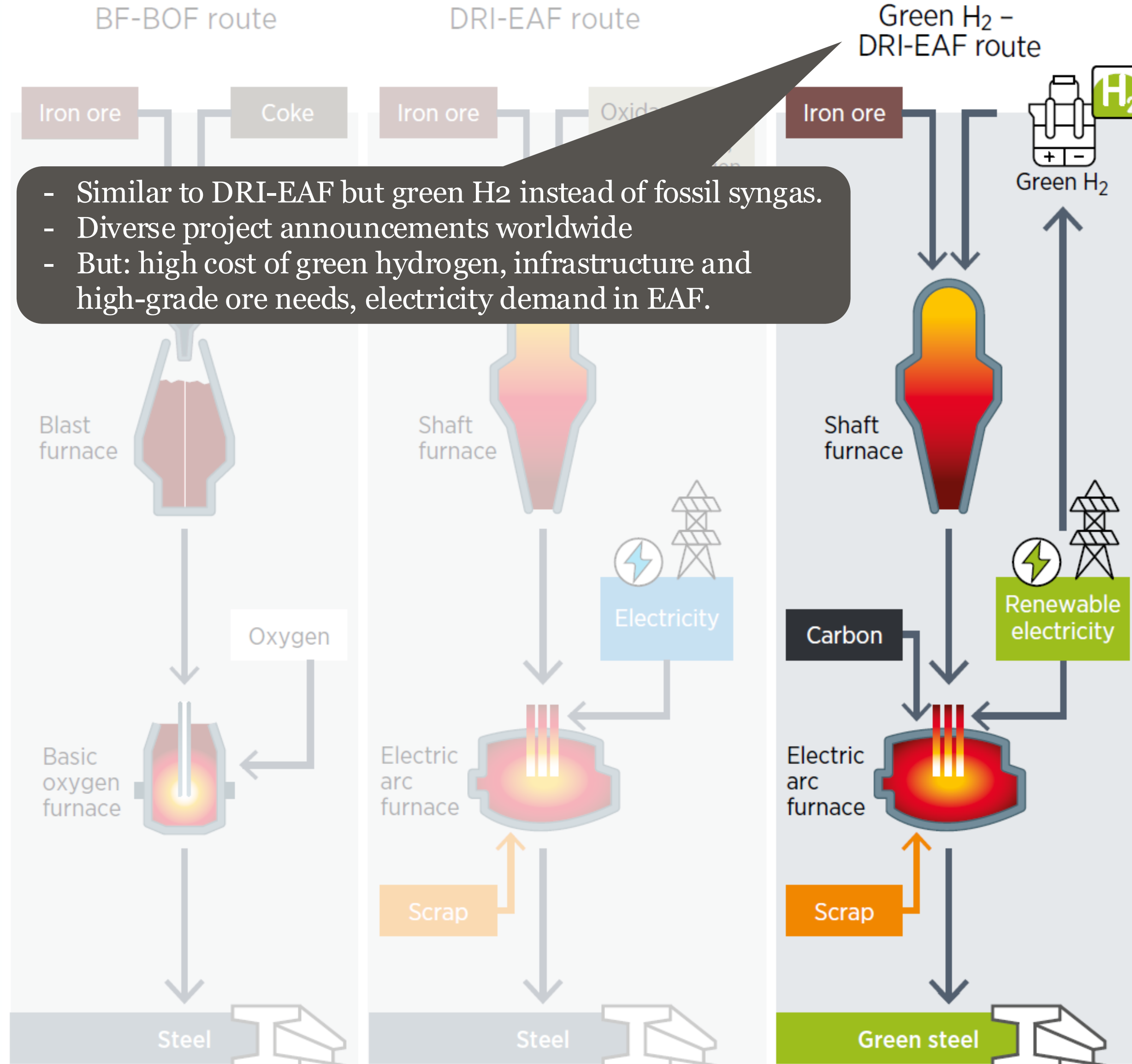
DRI-EAF

Hydrogen replaces CO/CH_4 as reductant

Reaction: $Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$

Only byproduct: Water vapor

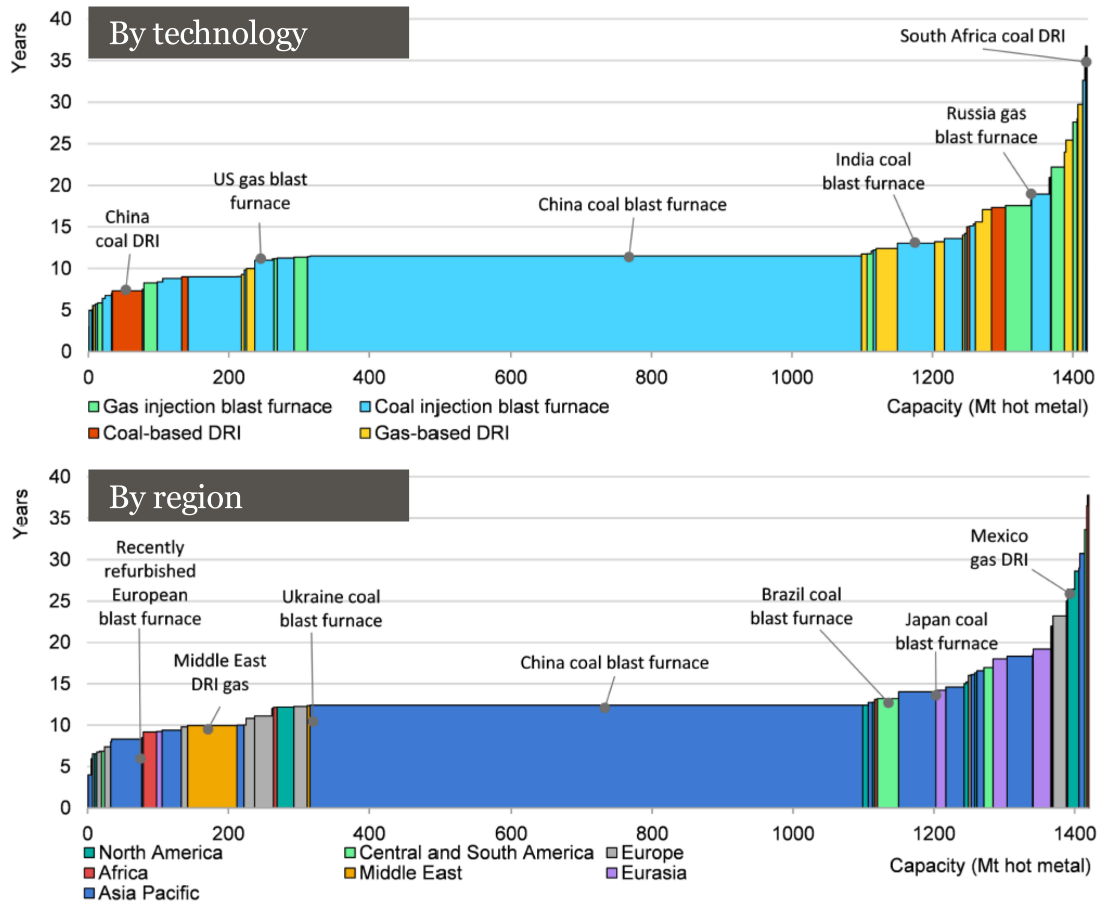
Paired with renewable electricity for near-zero emissions



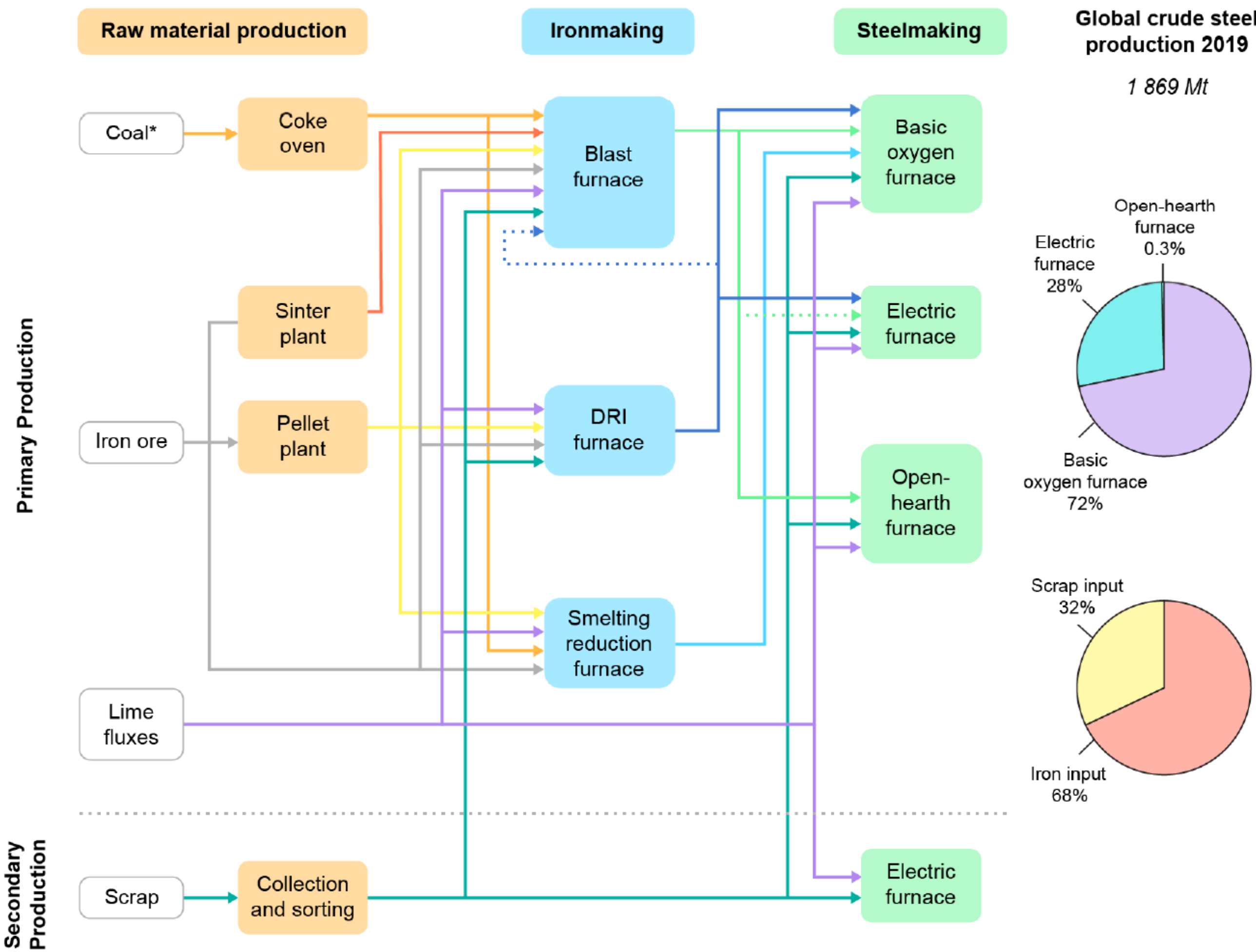
- Similar to DRI-EAF but green H2 instead of fossil syngas.
- Diverse project announcements worldwide
- But: high cost of green hydrogen, infrastructure and high-grade ore needs, electricity demand in EAF.

One issue of decarbonizing is long asset lifetimes! 40-50y

Figure 1.6 Geographic distribution and average age of main assets in the iron and steel sector by production route and region

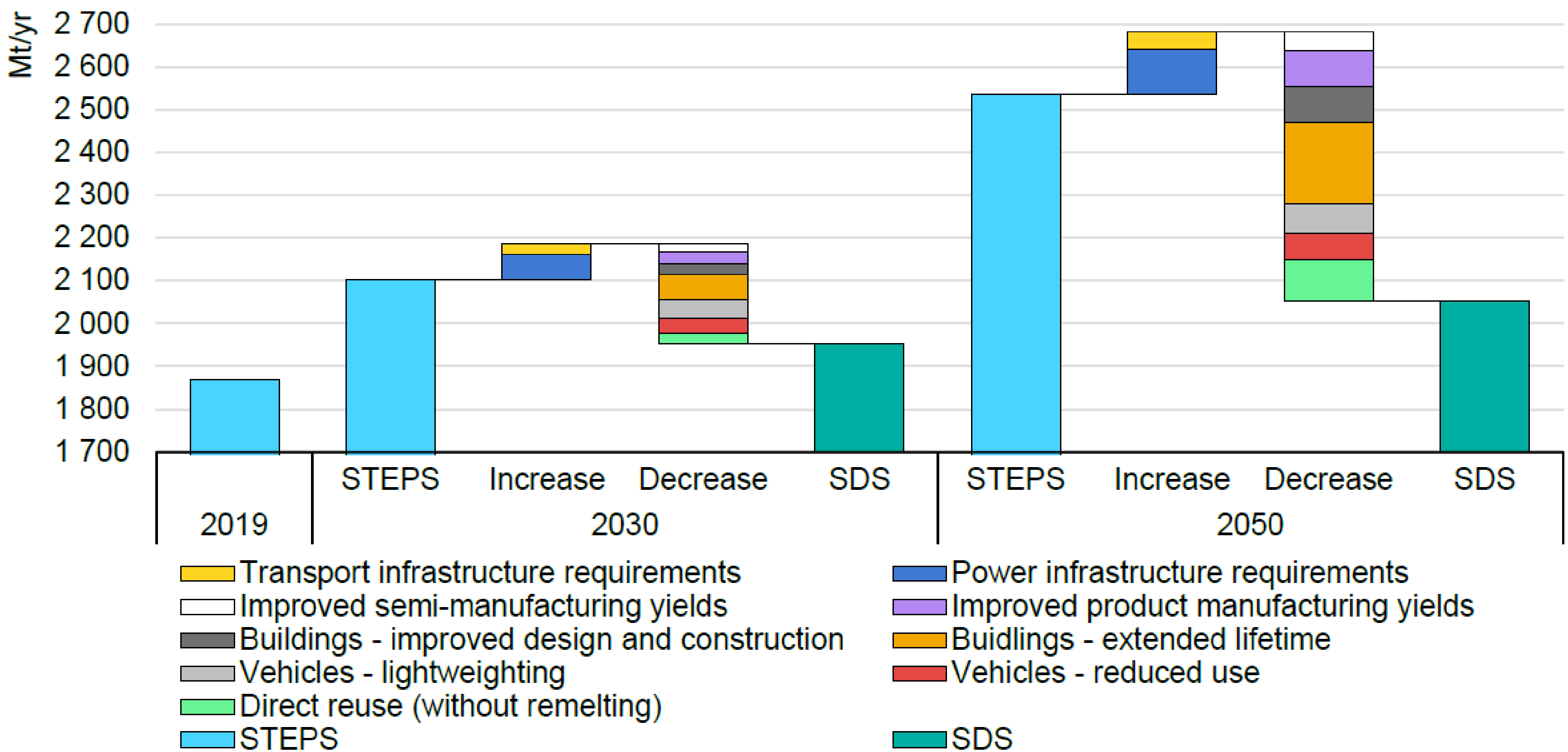


Ways of iron production flow chart



IEA 2020, Iron and Steel Technology Roadmap

Figure 2.3 The contribution of material efficiency strategies to reductions in global steel demand



Challenges for decarbonizing

1. Fossil carbon as fuel AND reducing agent => requires change in process
2. High-temperature coke ovens and sintering
3. Long asset lifetimes (~40–50 years)

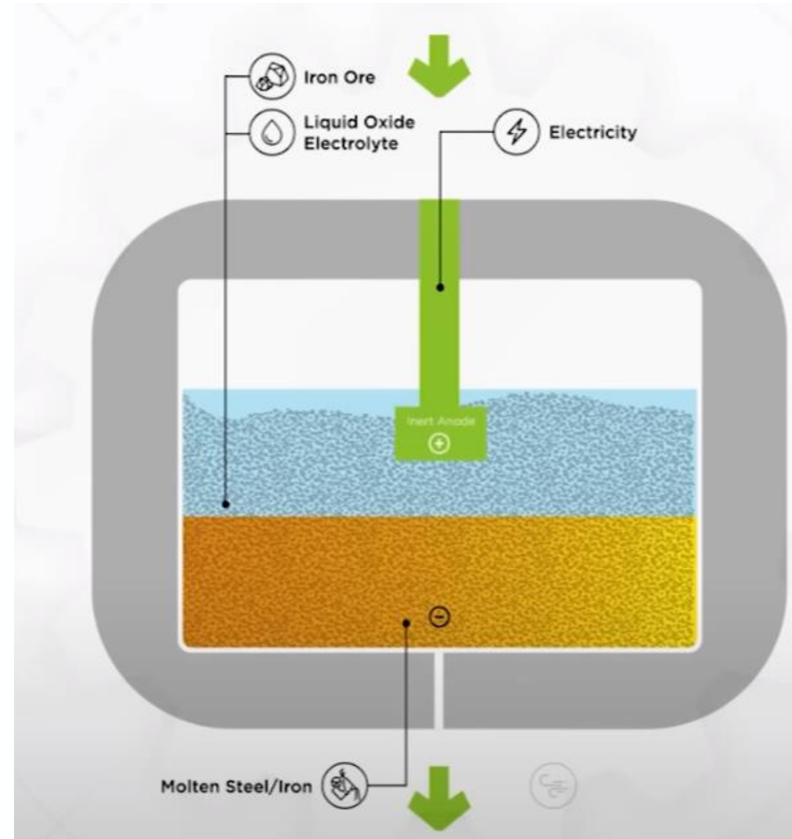
Emerging/niche technologies

Molten oxide electrolysis

From Sadoway at MIT (Boston Metals) – also called "electrowinning"

Thermo-electro-chemical process

- Needs heat and electricity
- 1500 C (challenging for intermittency)
- Very efficient
- Can do iron-ore to steel



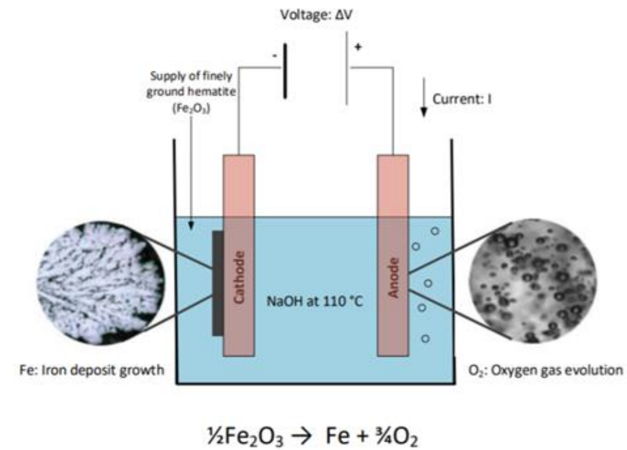
Low temperature electrolysis

Or called “electrowinning”

Two steps

- Dissolve iron-ore in acid (S) to make $FeSO_4$
- Then reduce electrochemically (electrolysis)

  $2FeSO_4 + 2H_2O \Rightarrow 2Fe + 2H_2SO_4 + O_2$
- Barriers:
  - hydrogen embrittlement
  - slow “batch” output



Electrowinning has actually existed for decades on a small scale to serve ultra-high-purity iron markets

- Aerospace & defence
- Electronics & semiconductors

Cement production

Cement production steps and emissions

The primary source of emissions in cement is actually not from burning fossil fuels

1. The Chemical Process: Calcination (~60% of emissions)

Limestone ($CaCO_3$) blasted with intense heat = quicklime (CaO) and CO_2

$$CaCO_3 \xrightarrow{heat} CaO + CO_2$$

2. The Energy Source: Combustion (~30% to 35% of emissions)

Heat required at ~1450°C –> traditionally requires burning fossil fuels

3. Misc (~5% of emissions)

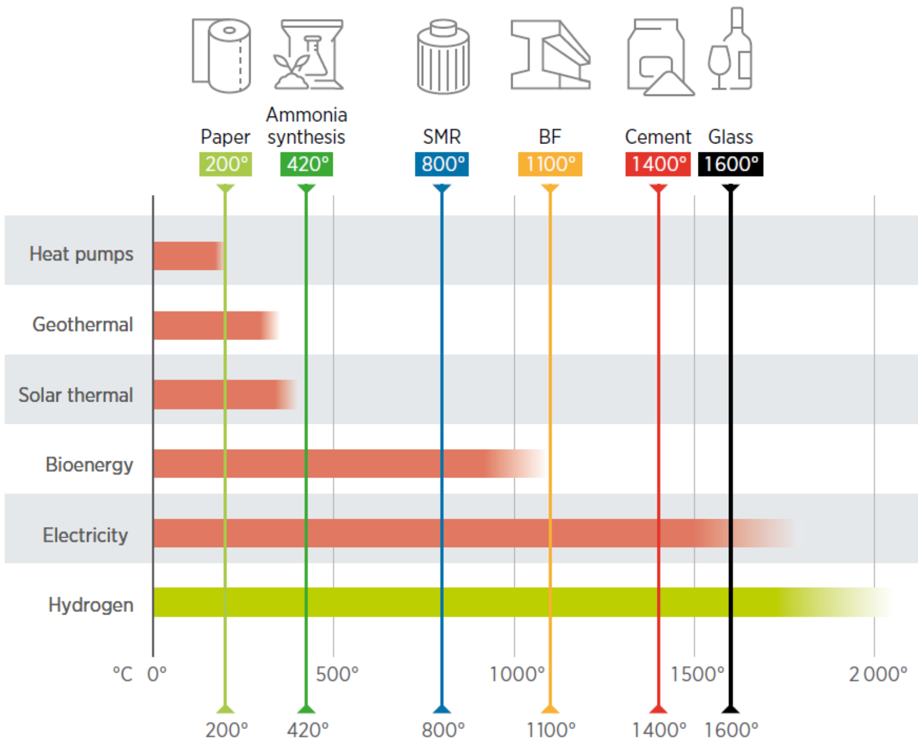
Crushers, grinders, and plant logistics – can be directly electrified

How to decarbonize cement?
(Discuss)

How to decarbonize cement production?

30-35% emissions from heat

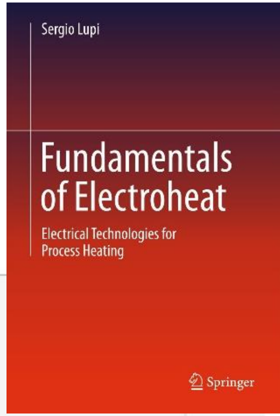
1. High-temp electric heat (plasma torches, resistive or inductive heating)
2. Use biomass, waste-derived fuels, or green hydrogen



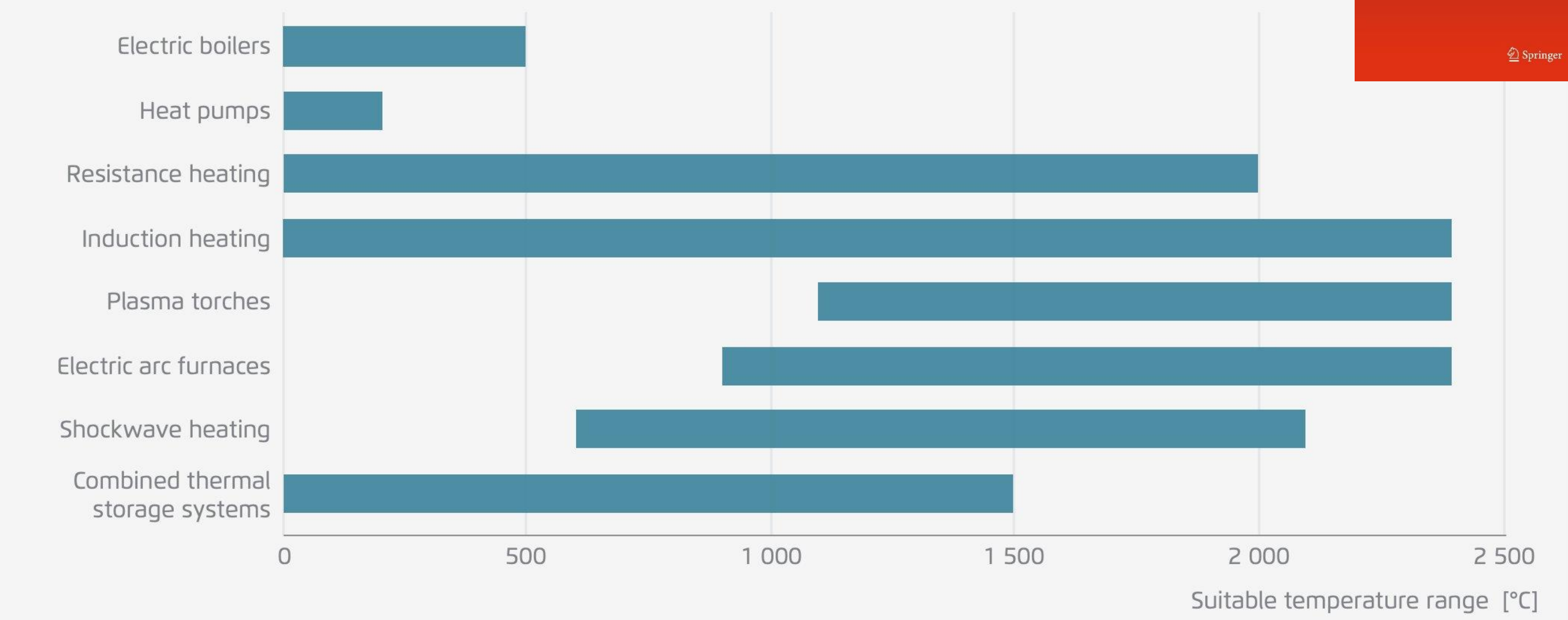
Fennell et al 2022: Going net zero for cement and steel. Commentary in Nature

Cement Sustainability Initiative 2016. Cement Industry. Energy and CO2 Performance: Getting the Numbers Right

Zooming into electric heat options



Potential development of suitable temperature range of electric heating technologies until 2035



Fraunhofer ISI (2024)

How to decarbonize cement production?

Process emissions (~60%)

some researchers call process emissions from cement "*unavoidable*"

1. Capture the CO_2
2. Reduce amount of cement needed (switch to timber)
3. Substitute a portion of limestone with "supplementary cementitious materials" (fly ash, limestone calcined clay cement)

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Concluding remarks

Common misconceptions on green steel

TLDR The technology we need already exists!

1. "If we recycled steel, we would not need any new steel"
2. "Using electric arc can't be done at the scale needed"
3. "Hydrogen for steel is unproven technology and might produce weak steel"
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1\. We already do recycle most of the scrap steel (cheaper, cleaner).

2\. We use electric arc in diverse industries, inc. scrap steel (25% of world production)

3\. While efforts are just ramping up, electrolyzers are modular and quickly scalable. The input of H_2 and carbon is controlled to be chemically equivalent to other steel production processes.

4\. While carbon is used, the amount is low and can be sourced from biogenic source. Overall carbon requirements are brought down to 0.025tCO2/ ton steel

UC

Timber!

Solves 3 issues

Less steel

Less cement

Captures CO_2

Study goals

Today, we want to understand

What are hard-to-abate sectors?

- Complexity beyond just “plugging in” green electricity
- Like steel and cement, about 7 and 5% of global emissions

Understanding decarbonization strategies for steel

- Currently, most steel comes from BF-BOF route. Tough to decarbonize.
- Second most used method is DRI-EAF. Can be decarbonized by switching to H_2
  - The required technologies all exist!
- Emerging technologies, perhaps a decade away: Molten oxide EL, and Low temperature EL
- More details in “Iron and Steel Technology Roadmap” from IEA

Understanding decarbonization strategies for cement

- ~35% from burning fuels for heat => renewable heat
- ~60% of the emissions come from the chemical reaction (!)
  - Capture CO_2,
  - Other materials (promising: LC),
  - Reduce need for cement (timber buildings).

I want to learn more

Recommended

- IRENA, 2022: "Green hydrogen for industry - a guide to policy making"
- Fennel et al 2022: Going net zero for cement and steel. Commentary in Nature. https://www.nature.com/articles/d41586-022-00758-4
- Engineering with Rosie, 2022 (Dec): "Revolutionizing Steel Production: The Groundbreaking Solution to Cut Emissions!" https://www.youtube.com/watch?v=KPAeuJvu7ps
- Engineering with Rosie, 2022: "Green Steel: Can We Make Steel Without CO2 Emissions?" https://www.youtube.com/watch?v=jWD2nI5RhpI

Reference

- IEA 2022 "Iron and Steel Technology Roadmap"
- Lupi, 2017: "Fundamentals of Electroheat". Springer
- DW, 2022: "How to fix the world's concrete problem" https://www.youtube.com/watch?v=dgVWz77lGCo